Additional Exercises 4

Gradient, Directional Derivatives & Higher-Order Partial Derivatives

Mirrors: List 5 | ★ Easy ★★ Medium ★★★ Hard

Exercise 1: Calculate the directional derivative

Recall: $D_{\hat{u}}f = \nabla f \cdot \hat{u}$ *where* $\hat{u}$ *is the unit vector.*

(a) ★ $f(x,y) = x^2 + xy - y^2$ at $(1,2)$ in direction $\bar{u} = (3,4)$.

(b) ★ $f(x,y,z) = xy + yz + zx$ at $A(2,1,3)$ in direction $\bar{a} = (3,4,12)$.

(c) ★ $f(x,y) = e^x \cos y$ at the origin in direction making angle $\pi/4$ with the x-axis.

(d) ★★ $f(x,y,z) = \ln(x^2 + y^2 + z^2)$ at $A(1,2,1)$ in direction $\bar{a} = (2,4,4)$.

(e) ★★ $f(x,y,z) = xyz$ at $A(2,4,2)$ in direction $\bar{u} = (4,2,-4)$.

(f) ★★ Find the directional derivative of $f(x,y) = x^2y$ at $P(1,1)$ in the direction of the unit vector making angle $\theta = \pi/3$ with the positive x-axis.

(g) ★★★ $f(x,y) = \begin{cases} \dfrac{x^4y^2}{x^4+y^2}, & (x,y) \neq (0,0), \\ 0, & (x,y) = (0,0). \end{cases}$ at $(0,0)$ in direction $\bar{u} = (1,1)$.
(Use the limit definition — the formula $\nabla f \cdot \hat{u}$ *may not apply here.)*

Exercise 2: Find the direction (if it exists) for the given rate of change

(a) ★ $f(x,y) = 3x + 4y$ at any point. Find the direction of steepest ascent and its rate.

(b) ★★ $f(x,y) = x^2y + y^3$ at $(1,-1)$. Find a unit vector $\hat{u}$ such that $D_{\hat{u}}f = 0$.

(c) ★★ $f(x,y,z) = xy + z$ at $(0,1,-2)$. Find $\hat{u}$ such that $D_{\hat{u}}f = 1$. Does a direction with $D_{\hat{u}}f = 20$ exist?

(d) ★★★ $f(x,y) = x^2y + y^3$ at $(1,-1)$. Find $\hat{u}$ such that $D_{\hat{u}}f = 20$, or show no such direction exists.

Exercise 3: Find the gradient of the scalar field at point M_0

(a) ★ $u = x + y + 2xy$ at $M_0(1,1)$

(b) ★ $u = x^2 + y^2 + z^2$ at $M_0(2,0,0)$

(c) ★★ $u = \ln(x^2 + y^2 + z^2)$ at $M_0(1,1,1)$

(d) ★★ $u = x^2 e^{y-z}$ at $M_0(2,1,1)$

(e) ★★ $u = \arctan\left(\frac{y}{x}\right) + \arctan\left(\frac{x}{y}\right)$. Compute ∇u and explain the result.

(f) ★★★ $u = \arctan(xyz) + \operatorname{arccot}(xyz)$. Compute ∇u and explain the result.

(g) ★★★ $u = f(r)$ where $r = \sqrt{x^2 + y^2 + z^2}$, f differentiable. Express ∇u in terms of $f'(r)$ and $\bar{r} = (x, y, z)$.

Exercise 4: Second-order partial derivatives

Verify Schwarz's theorem where asked: $f_{xy} = f_{yx}$.

(a) ★ $z = \sin(xy)$. Find $\dfrac{\partial^2 z}{\partial x\, \partial y^2}$.

(b) ★ $z = x^3 y + xy^3$. Verify $f_{xy} = f_{yx}$.

(c) ★★ $z = x^y$. Verify $f_{xy} = f_{yx}$.

(d) ★★ $u = \arctan(y/x)$. Verify the Laplace equation: $\dfrac{\partial^2 u}{\partial x^2} + \dfrac{\partial^2 u}{\partial y^2} = 0$.

(e) ★★ $u = e^x(x\cos y - y\sin y)$. Verify $u_{xx} + u_{yy} = 0$.

(f) ★★★ $u = \ln\dfrac{1}{r}$, $r = \sqrt{(x-a)^2 + (y-b)^2}$. Verify that u satisfies the Laplace equation for $(x, y) \neq (a, b)$.

Exercise 5: Find the second-order differential d^2u

Recall: $d^2u = u_{xx}(dx)^2 + 2u_{xy}\, dx\, dy + u_{yy}(dy)^2$.

(a) ★ $u = x^2 y^3$

(b) ★ $u = e^{xy}$

(c) ★★ $u = x^y$

(d) ★★ $u = x\ln(xy)$

(e) ★★ $u = \sin(x + y) + \cos(x - y)$

(f) ★★★ $u = f(x^2 - y^2)$, f twice-differentiable. Express d^2u in terms of f' and f''.

(g) ★★★ $u = \ln\sqrt{x^2 + y^2}$. Compute d^2u and verify $u_{xx} + u_{yy} = 0$.

★ *Direct formula* ★★ *Gradient properties or Schwarz* ★★★ *Definition from first principles or abstract function.*